

AURCHI CHOWDHURY

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EDUCATION

Bangladesh University of Engineering and Technology <i>Bachelor of Engineering in Computer Science and Engineering CGPA: 3.88/4.00</i>	Nov 2022 – Present <i>Dhaka, Bangladesh</i>
Viqarunnisa Noon School and College <i>High School GPA: 5.00/5.00 Board Scholarship</i>	Jan 2009 – Dec 2021 <i>Dhaka, Bangladesh</i>

INTERESTS

Natural Language Processing · Multimodal AI · Computer Vision · Machine Learning · Mechanistic Interpretability

PUBLICATIONS

🌐 Causal Localization of the English Pivot in LLaVA: Mechanistic VLM Analysis and Training-Free Multilingual Steering <i>1st Workshop on Multilinguality in the Era of Large Language Models (MeLLM), ACL 2026 Archival</i>	Jul 2026 <i>San Diego, CA</i>
🌐 Where Do Affordances Crystallize? Video vs. Image Self-Supervised Encoders for Embodied Perception <i>7th Annual Embodied AI (EAI) Workshop, CVPR 2026 Non-archival Poster</i>	Jun 2026 <i>Denver, CO</i>

WORK EXPERIENCE

Undergraduate Research Intern Progya: Adaptive Tutoring Platform Dept. of CSE, BUET (UGC-funded)	On-site Nov 2025 – May 2026
- Developing an adaptive tutoring platform using Bayesian Knowledge Tracing (BKT), building learning pathways aligned with the BCS exam syllabus. - Supervisor: Md. Saifur Rahman, Md. Shariful Islam Bhuiyan	

PROJECTS

🌐 MemoryLane <i>PyTorch, ConvNeXt, U-Net, PatchGAN, VGG</i>	Apr 2026
- Framed grayscale-to-color mapping as a conditional image generation problem in the CIELab color space, building a ConvNeXt-Base encoder – U-Net decoder architecture. - Incorporated a PatchGAN discriminator and mitigated the gray-mean saturation bias endemic to deep colorization models via colorfulness-filtered training on ~518K Places365 images. - Designed a composite perceptual + adversarial loss (VGG feature matching + PatchGAN) to balance global structure with local texture fidelity. - Surpassed ground-truth colorfulness (40.9 vs. 36.9) and generalized to real, out-of-distribution family photographs.	
🌐 Scramble <i>ATMega32, SG90 servo motors, HC-05 Bluetooth module, Kotlin</i>	Jul 2025
- Developed a Fully Automated Rubik's Cube Solver capable of generating solutions in a maximum of 20 moves. - Used two ATMega32 microcontrollers, eight SG90 servo motors and an HC-05 Bluetooth module.	
📺 Project Demonstration	
🌐 On the Road <i>Node, Express, React, PostgreSQL</i>	Mar 2024
Built a full-stack travel planning platform covering accommodations, dining, attractions, and tour guides.	

ACHIEVEMENTS

🌐 1st Runner Up, ResearchSpark 3 Minute Thesis Competition BUET	Apr 2026
🌐 1st Runner Up, HULT Prize at BUET	Feb 2026
🌐 2nd Runners Up, Research Poster Presentation BUET CSE Fest 2024	Oct 2024